

Computer Logic and Digital Circuit Design (PHYS306/COSC330): Unit 3

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Summary

Unit 3 Summary

Reading: chapter 8, 9

1. Shift-Registers

- Shift register I/O
- Load, shift, and bidirectional functions
- Shift register counters
- Application examples

2. Counters

- Finite State Machines (FSMs)
- Asynchronous/Synchronous counters
- Up/Down functions
- Cascaded counters
- Counter decoding
- Counter Applications

Shift registers

Shift register I/O

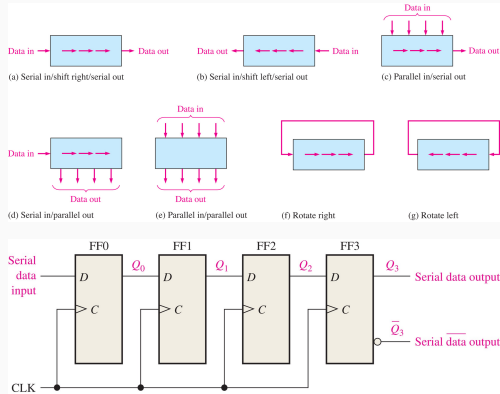


Figure 1: Different forms of shift register I/O. Data in/out can be serial or parallel, and leftward or rightward. Registers can be circular, rotating data.

Shift register functions: load/shift

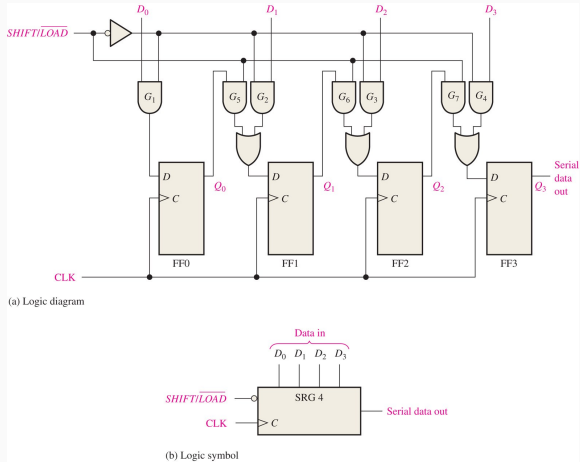


Figure 2: A 4-bit shift register from D flip-flops, with shift/load function.

Shift register functions: load/shift

TABLE 8-1

Shifting a 4-bit code into the shift register in Figure 8-3. Data bits are indicated by a beige screen.

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	0	0	0	0
1	0	0	0	0
2	1	0	0	0
3	0	1	0	0
4	1	0	1	0

TABLE 8-2

Shifting a 4-bit code out of the shift register in Figure 8-3. Data bits are indicated by a beige screen.

CLK	FF0 (Q_0)	FF1 (Q_1)	FF2 (Q_2)	FF3 (Q_3)
Initial	1	0	1	0
5	0	1	0	1
6	0	0	1	0
7	0	0	0	1
8	0	0	0	0

Figure 3: Shifting a 4-bit sequence into a 4-bit shift-register.

- Note the direction of data flow.
- Data bits are initially RESET.
- Positive edge CLK pulses advance data.

Shift register functions: bidirectional

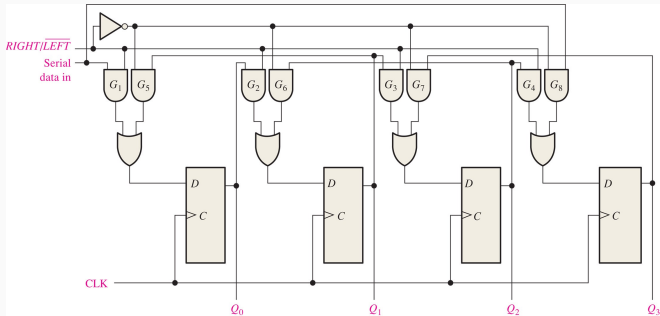


Figure 4: A 4-bit shift register from D flip-flops, with bidirectional function.

- Note *backward connections*.
- Steering gates load rightward or leftward data.

Shift register functions: bidirectional

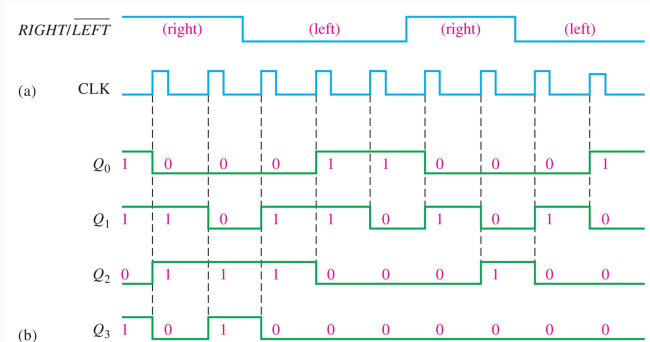


Figure 5: A 4-bit shift register from D flip-flops, with bidirectional function.

- Note *backward connections*.
- Steering gates load rightward or leftward data.

Shift register counters: Johnson counters

TABLE 8-3

Four-bit Johnson sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3
0	0	0	0	0
1	1	0	0	0
2	1	1	0	0
3	1	1	1	0
4	1	1	1	1
5	0	1	1	1
6	0	0	1	1
7	0	0	0	1

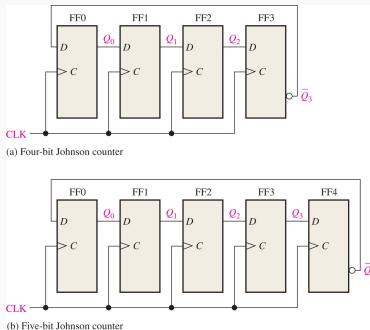


Figure 6: (Left) A 4-bit Johnson counter state sequence. (Right) Circuit diagram of Johnson counters.

- Note the output requires a decoder.
- Final output is active LOW.

Shift register counters: ring counters

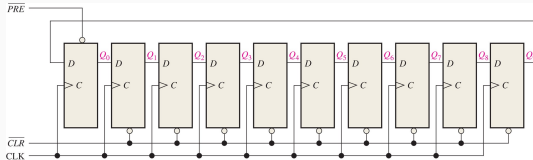


TABLE 8-5

Ten-bit ring counter sequence.

Clock Pulse	Q_0	Q_1	Q_2	Q_3	Q_4	Q_5	Q_6	Q_7	Q_8	Q_9
0	1	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
2	0	0	1	0	0	0	0	0	0	0
3	0	0	0	1	0	0	0	0	0	0
4	0	0	0	0	1	0	0	0	0	0
5	0	0	0	0	0	1	0	0	0	0
6	0	0	0	0	0	0	1	0	0	0
7	0	0	0	0	0	0	0	1	0	0
8	0	0	0	0	0	0	0	0	1	0
9	0	0	0	0	0	0	0	0	0	1

Figure 7: (Top) A 10-bit ring counter diagram. (Bottom) States corresponding to the ring counter. Note the output requires a decoder.

Shift register applications: time-delay

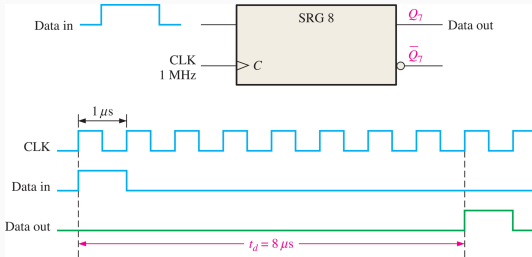


Figure 8: (Top) An 8-bit shift register as a time-delay. (Bottom) Corresponding timing diagram assuming 1 MHz CLK frequency.

Shift register applications: serial to parallel data conversion

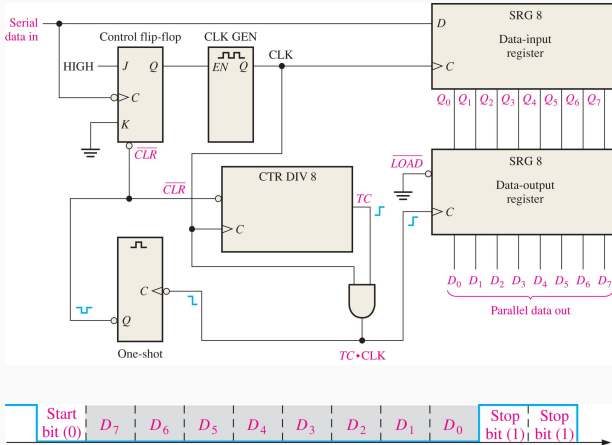


Figure 9: (Top) Serial to parallel data converter circuit diagram. (Bottom) Concept of start and stop bits.

Shift register applications: serial to parallel data conversion

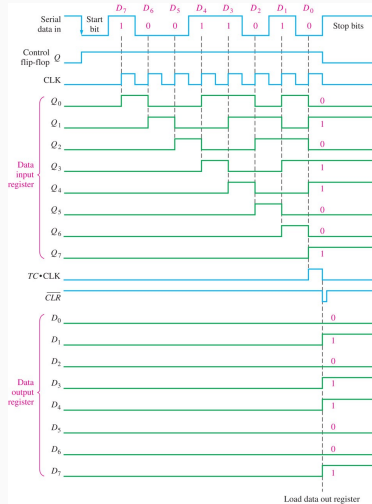


Figure 10: Timing diagram of the converter.

Counters

Counters: Finite State Machines

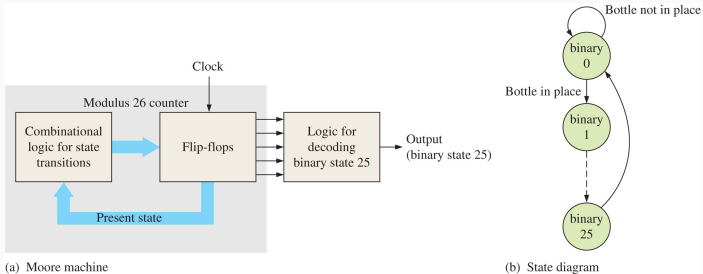


Figure 11: Example of a *Moore machine*.

Counters: Finite State Machines

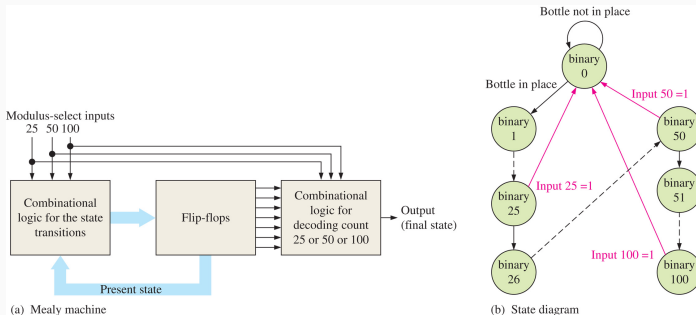


Figure 12: Example of a *Mealy machine*.

Counters: Asynchronous counters

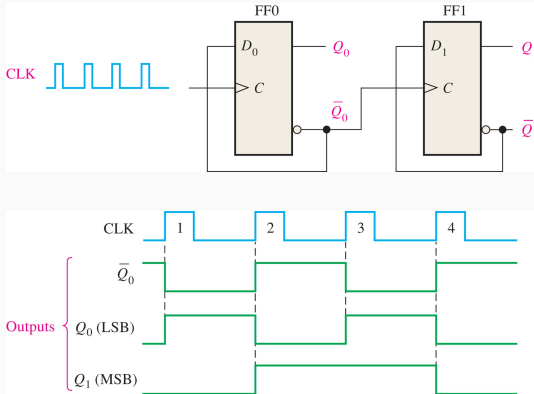


Figure 13: (Top) 2-bit asynchronous counter circuit diagram. (Bottom) Corresponding timing diagram.

Counters: Synchronous counters

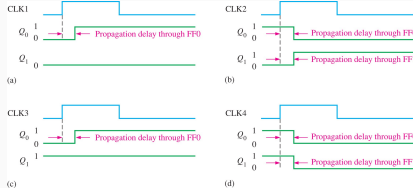
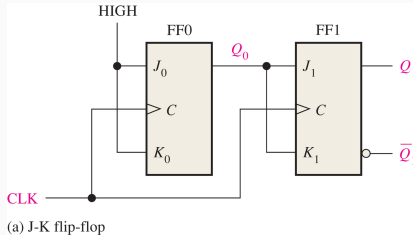


Figure 14: (Top) A 2-bit synchronous counter based on J-K flip flops. (Bottom) Timing detail for the 2-bit synchronous counter.

Counters: Synchronous counters

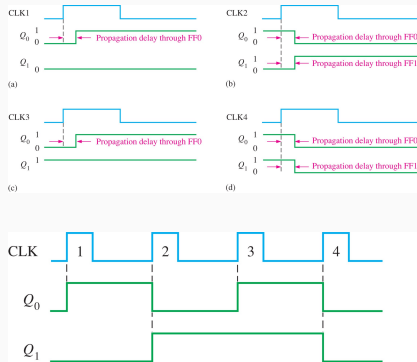


Figure 15: (Top) Timing detail for the 2-bit synchronous counter. (Bottom) Full timing diagram.

Counters: Synchronous counters

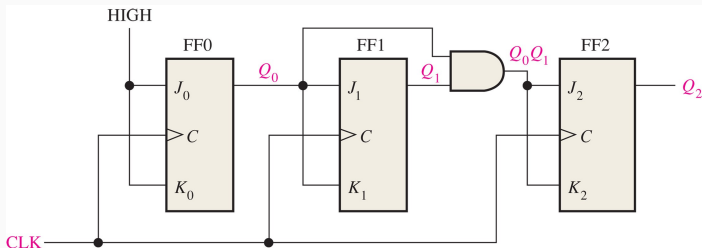


Figure 16: A 3-bit synchronous counter requires a steering gate.

Counters: Synchronous counters

TABLE 9-4

Summary of the analysis of the counter in Figure 9-15.

Clock Pulse	Outputs			J-K Inputs						At the Next Clock Pulse		
	Q_2	Q_1	Q_0	J_2	K_2	J_1	K_1	J_0	K_0	FF2	FF1	FF0
Initially	0	0	0	0	0	0	0	1	1	NC*	NC	Toggle
1	0	0	1	0	0	1	1	1	1	NC	Toggle	Toggle
2	0	1	0	0	0	0	0	1	1	NC	NC	Toggle
3	0	1	1	1	1	1	1	1	1	Toggle	Toggle	Toggle
4	1	0	0	0	0	0	0	1	1	NC	NC	Toggle
5	1	0	1	0	0	1	1	1	1	NC	Toggle	Toggle
6	1	1	0	0	0	0	0	1	1	NC	NC	Toggle
7	1	1	1	1	1	1	1	1	1	Toggle	Toggle	Toggle
Counter recycles back to 000.												

*NC indicates *No Change*.

Figure 17: The internal states, and outputs for a 3-bit synchronous counter from J-K flip-flops.

Conclusion

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